

AP CSP | Lesson 02: Computing Innovations

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 questions in about 40 minutes. For Q5 select exactly TWO answers. Name actual input/output data and explain the program boundary. Written responses use classroom rubrics.

1. A room device sends 00011010 as an unsigned binary integer. What decimal value does the program receive? [1 mark]

A. 24 B. 25 C. 26 D. 28

Answer: _____

2. A museum needs distinct fixed-length binary codes for 35 rooms. No codes are reserved. What is the minimum number of bits per code? [1 mark]

A. 5 B. 6 C. 7 D. 35

Answer: _____

3. A program matches volunteer tutors with learners who request help. Which statement best describes a plausible purpose? [1 mark]

A. It reads subjects and available times. B. It displays a list of matches.
C. It compares stored profiles. D. It helps learners find suitable tutoring support.

Answer: _____

4. Which example clearly uses a program as an essential part of its function? [1 mark]

A. A spring-driven kitchen timer with no electronics B. A paper map marked by a traveler
C. Software that combines recorded sounds into music D. A wooden ruler used to measure a shelf

Answer: _____

5. A garden program reads soil-moisture measurements and a target moisture level entered by a user. It then sends a watering command. Which TWO are inputs? Select exactly TWO answers. [1 mark]

A. The measured soil-moisture data B. The watering command sent to a controller
C. The target moisture level entered by the user D. The instructions that compare the two levels

Answer: _____

6. A counter displays 4. A button click increases the stored count by one and displays 5. Later, the same button click displays 9. Which explanation fits this behavior? [1 mark]

A. A click cannot be program input. B. The output depends on both the click and the previous count.
C. The display device changes the arithmetic rule. D. The same click must always produce the same displayed number.

Answer: _____

AP CSP | Lesson 02: Computing Innovations

Written responses | Questions 7-9

7. [4 marks]

A library app reads a topic selected by a reader, finds matching books in a catalog, and displays titles.

- (a) State a plausible purpose naming a user and goal.
- (b) Describe its function.
- (c) Identify one input and one output, naming the actual data.

8. [4 marks]

A forecast program reads a file of temperatures 18, 21 and 24. It calculates their mean and writes "Mean: 21" to a results file.

- (a) Identify the input data and its source.
- (b) Describe the processing.
- (c) Identify the output.
- (d) Explain why the saved result is still output without a screen display.

9. [4 marks]

A sensor program sends "occupied" to a room-finder program. The room finder removes that room from its available-room list and displays the revised list.

- (a) Classify the message relative to each program.
- (b) Identify processing performed by the room finder.
- (c) Identify its final displayed output.

AP CSP | Lesson 02: Computing Innovations

Written responses | Questions 10-12

10. [4 marks]

An app stores a tally. Each click on “Add one” increases the tally by one and displays it.

- (a) Identify the event input.
- (b) State the displayed result if the tally was 2, and if it was 8.
- (c) Explain why the same click can produce different outputs.
- (d) Is the new tally a program instruction or data?

11. [4 marks]

Students want a room finder for group study.

- (a) Write one interview question that could reveal a user need or constraint.
- (b) Name one input data item and one useful output.
- (c) Write one requirement with an observable result. State enough detail to check it.

12. [4 marks]

A room finder must show all available rooms with enough seats. A has 2 seats and is available; B has 6 and is available; C has 8 and is unavailable.

- (a) Give the result for a group of 2, then a group of 7.
- (b) For 7 students, the app displays C. Explain the error.
- (c) Propose clear output when no room meets the requirement.
